Detecting AI-Generated Student Essays on Unseen Topics

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Abstract

Teachers worry that students hand in essays written by large language models. This report asks how often text classifiers wrongly flag a student's own essay when the topic is new. I compare two classical classifiers with two fine-tuned transformer encoders on about 45,000 essays. On a random split the task looks solved. On unseen topics, the answer depends on the topic. For three of the four models, most false accusations come from one of the three test topics, where up to 16% of the student essays are flagged, while ModernBERT flags only 0.9% overall. For the classical models, holding out each of the 15 topics in turn confirms that this topic is the worst, though not the only bad one.

1 Introduction

Many teachers worry that students hand in texts written by AI. My master's project studies how NTNU teachers use and avoid AI, and this worry comes up there too. This project looks at the technical side: can a classifier tell student essays from essays written by large language models (LLMs), and how often is it wrong?

For a teacher, the two possible errors are not equally bad: flagging a student who wrote their own essay is a false accusation. I therefore report the false positive rate, the share of student essays flagged as AI, next to the usual metrics.

The research questions are:

RQ1 How well do TF-IDF with logistic regression and a linear SVM separate student and AI essays?

RQ2 Do fine-tuned transformers do better, and does ModernBERT improve on DistilBERT?

RQ3 How much does performance drop on new essay topics, as when a teacher writes a new assignment?

RQ4 Which essays do the models get wrong, and what do they have in common?

Sections 2 and 3 cover background and earlier work, Section 4 the data and method, Section 5 the results, and Section 6 what they mean for teachers.

2 Background

TF-IDF weights each word or word pair (n-gram) in an essay by how often it occurs there, discounted by how many essays contain it (Spärck Jones, 1972). With sublinear term frequency, as used here, the weight of n-gram t in essay d is

$$w_{t,d} = \left(1 + \ln \mathrm{tf}_{t,d}\right) \cdot \left(\ln \frac{1+N}{1+\mathrm{df}_t} + 1\right), \tag{1}$$

where $\mathrm{tf}_{t,d}$ is the count of t in d, N the number of training essays and df_t the number of essays containing t. Logistic regression and a linear support vector machine (SVM) (Cortes and Vapnik, 1995) both learn

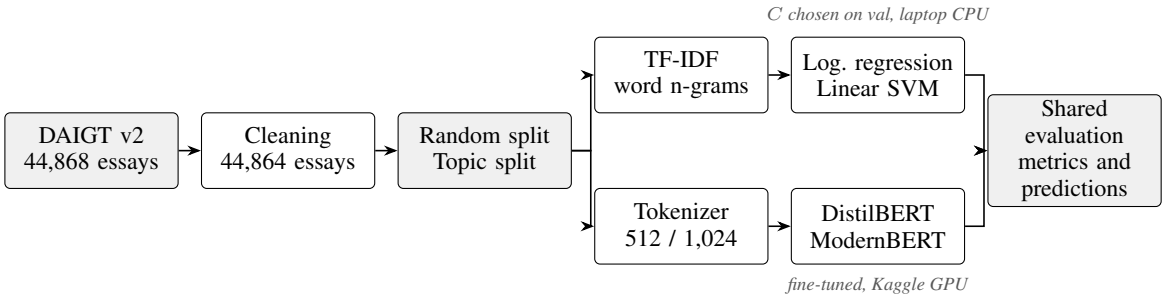


Figure 1: The experimental pipeline. All four models are trained on both splits and scored by the same evaluation code. Each run saves its metrics and one prediction per test essay, which the error analysis reads without retraining.

one weight per n-gram, with a parameter C for regularisation. SVMs have long been strong baselines for text classification (Joachims, 1998).

Both transformers are BERT-style encoders (Devlin et al., 2019; Vaswani et al., 2017), fine-tuned with a new classification head (Jurafsky and Martin, 2026). DistilBERT (Sanh et al., 2019) is a compressed BERT that is 40% smaller and 60% faster while keeping 97% of its language understanding. It reads at most 512 tokens. ModernBERT (Warner et al., 2025) is a recent encoder with a context length of 8,192 tokens, trained on 2 trillion tokens. They differ in how they pool the token vectors $\mathbf{h}_1, \ldots, \mathbf{h}_n$ into one vector (Jurafsky and Martin, 2026):

$$\hat{y} = \mathrm{softmax}(W\mathbf{h}), \qquad \mathbf{h}_{\text{DistilBERT}} = \mathbf{h}_{\text{[CLS]}}, \qquad \mathbf{h}_{\text{ModernBERT}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{h}_i. \tag{2}$$

3 Related Work

Zero-shot detectors such as DetectGPT (Mitchell et al., 2023) use the probabilities a language model assigns to a text, without training a classifier. Supervised detectors learn from labelled examples instead: a fine-tuned RoBERTa model (Liu et al., 2019) detected text from the largest GPT-2 model (Radford et al., 2019) with about 95% accuracy (Solaiman et al., 2019).

Alikhanov et al. (2026) combined DAIGT v2 with the HC3 dataset and found that a random split gave too optimistic results, even for logistic regression, because the model learned topic vocabulary. They report no numbers for this, and Table 4 quantifies it.

The errors can also fall unevenly. On average, seven widely used detectors flagged more than half of a set of TOEFL essays written by non-native English speakers as AI-generated (Liang et al., 2023).

4 Data and Method

This section describes the data, the two splits and the four models, and the pipeline is shown in Figure 1.

4.1 Data

I use the DAIGT V2 Train Dataset (thedrcat, 2023), made for the Kaggle competition LLM – Detect AI Generated Text (The Learning Agency Lab, 2023). Each essay answers one of 15 prompts (topics), and a source column records where it came from. According to that column, 95% of the student essays come from the PERSUADE corpus, written by students in grades 6 to 12 (Crossley et al., 2023). The AI essays come from 16 sources, including Mistral, Llama, Falcon, PaLM, Cohere, Claude and GPT models.

After dropping missing, empty and duplicate texts, 44,864 of the 44,868 essays remain, and 39.0% of them are AI-generated.

4.2 Two Splits

The *random split* (Table 1) is a stratified 80/10/10 split, so all 15 topics appear in every part. The *topic split* holds out whole topics. The topics were shuffled with a fixed seed before any results were seen,

Table 1: The two splits after cleaning

Split	Part	Essays	AI share	Topics
Random	Train	35,890	39.0%	15
	Validation	4,487	39.0%	15
	Test	4,487	39.0%	15
Topic	Train	34,334	44.4%	10
	Validation	3,988	26.9%	2
	Test	6,542	17.9%	3

Table 2: Model settings. The transformers use AdamW, the Trainer's default optimiser

Classical models (scikit-learn, LIBLINEAR solver)		
Features	Word unigrams and bigrams, lower-cased, sublinear term frequency	
Vocabulary	N-grams in at least 2 essays, at most 200,000	
Classifiers	Logistic regression, linear SVM	
C	Best of $\{0.1, 1, 10\}$ by validation macro-F1	
Chosen C	Random split: 10 for both. Topic split: 1 (log. regression), 0.1 (SVM)	
Score	Signed distance from the decision boundary	
Transformers	DistilBERT	ModernBERT
Model	`distilbert-base-uncased`	`answerdotai/ModernBERT-base`
Parameters	66M	149M
Maximum input length	512 tokens	1,024 tokens
Pooling	[CLS] token	Mean of all tokens
Learning rate	$2 \cdot 10^{-5}$	$2 \cdot 10^{-5}$
Epochs	3	3
Batch size $\times$ accumulation	16×1	8×2
Weight decay	0.01	0.01
Warm-up	10% of steps	10% of steps
Precision	fp16	fp16
Checkpoint kept	Best validation macro-F1	Best validation macro-F1
Score	Softmax probability of AI	Softmax probability of AI

and the first three became the test set: *Driverless cars*, *Exploring Venus* and *Grades for extracurricular activities*. The next two became the validation set: *"A Cowboy Who Rode the Waves"* and *Community service*.

4.3 Models

Table 2 lists all model settings. The two classical models use the same TF-IDF features, fitted on the training essays only, so they differ only in the classifier. Both come from scikit-learn (Pedregosa et al., 2011).

The transformers are fine-tuned with Hugging Face Transformers (Wolf et al., 2020). Longer essays are cut, so the model only sees the start of the essay. In the topic test set, this happens to 22.7% of the essays for DistilBERT and 0.3% for ModernBERT. The settings are within the ranges recommended for fine-tuning BERT (Devlin et al., 2019), but they are not tuned.

5 Experiments and Results

This section defines the metrics, then gives the results overall, per test topic and over all 15 topics.

5.1 Experimental Setup

All runs use the seed 42. Settings are chosen on the validation set, and the test sets are only used for the final scores. Code and results are at https://github.com/johanneslyngaas/ai-essay-detection.

Table 3: Metrics. AI is the positive class

Metric	Meaning
AI recall	Share of AI essays that are caught, Equation 3
FPR	Share of student essays flagged as AI, Equation 3
ROC-AUC	Probability that a random AI essay gets a higher score than a random student essay. Does not depend on the threshold
Recall@1%	Best AI recall at an FPR of at most 1%, Equation 4

Table 4: Test results. Time is in seconds: the classical models ran on a laptop CPU and the transformers on one Kaggle GPU (T4), so the two are not comparable. Time includes feature extraction or tokenisation, model selection and test prediction

Split	Model	Acc.	Macro-F1	AI recall	FPR	ROC-AUC	Recall@1%	Time
Random	Log. regression	0.995	0.995	0.988	0.000	1.000	0.999	35
	Linear SVM	0.997	0.997	0.992	0.000	1.000	0.999	32
	DistilBERT	0.996	0.996	0.995	0.003	1.000	0.997	1,574
	ModernBERT	0.998	0.998	0.996	0.000	1.000	0.998	8,243
Topic	Log. regression	0.950	0.921	0.973	0.055	0.992	0.943	30
	Linear SVM	0.954	0.927	0.976	0.051	0.994	0.955	34
	DistilBERT	0.959	0.935	0.994	0.049	0.999	0.986	1,469
	ModernBERT	0.992	0.986	0.996	0.009	0.999	0.996	8,079

Table 3 defines the metrics. With a flagged student essay counted as a false positive (FP),

$$\text{FPR} = \frac{FP}{FP + TN}, \qquad \text{Recall} = \frac{TP}{TP + FN}. \tag{3}$$

Both depend on the threshold t on the model's score. Recall@1% is the best recall any threshold reaches while flagging at most 1% of the student essays:

$$\text{Recall@1\%} = \max_{t} \text{Recall}(t) \quad \text{subject to} \quad \text{FPR}(t) \leq 0.01. \tag{4}$$

Since t is chosen on the test scores, this is an upper limit, not what a deployed detector would achieve. Accuracy and F1 depend on the class balance, which differs between the test sets, so the splits are compared on AI recall, FPR, ROC-AUC and Recall@1%. For the topic split, 95% confidence intervals come from 1,000 bootstrap resamples of the test essays.

5.2 Experimental Results

Table 4 shows the results of the eight runs, one per model and split.

Random split. All four models are close to perfect: none flags more than nine of the 2,737 student essays.

Topic split. On unseen topics, the ranking is still very good, with ROC-AUC between 0.992 and 0.999, and every model catches at least 97% of the AI essays. The FPR, however, rises to 5.5% for logistic regression, 5.1% for the SVM and 4.9% for DistilBERT. Their confidence intervals overlap and together span 4.3–6.1%. ModernBERT flags only 0.9% (0.6–1.1%): 48 student essays against 264 for DistilBERT. Its AI recall is not clearly higher than DistilBERT's: 99.6% (99.2–99.9%) against 99.4% (98.9–99.8%).

Table 5: Errors on the topic split per test topic: student essays flagged as AI (false positives) and AI essays missed. LR is logistic regression, DB is DistilBERT and MB is ModernBERT

	Essays		False positives				Misses			
Topic	Student	AI	LR	SVM	DB	MB	LR	SVM	DB	MB
Driverless cars	1,886	364	2	0	13	0	16	15	5	4
Exploring Venus	1,862	314	**293**	**273**	**196**	17	0	0	0	0
Grades for extracurricular activities	1,626	490	2	2	55	31	15	13	2	1
All three topics	5,374	1,168	297	275	264	48	31	28	7	5

Table 6: Leave-one-topic-out results for the classical models. Each topic is held out in turn, and the models are trained on the other 14 with C fixed to the topic-split values. S marks the seven source-based prompts, where students write about a given text, and I the eight independent ones (from the `RDizzl3_seven` column in the dataset, which marks the seven prompts of the Kaggle competition, where students wrote about one or more source texts (The Learning Agency Lab, 2023)). Sorted by the FPR of logistic regression (LR)

			FPR		AI recall	
Held-out topic	Type	Student essays	LR	SVM	LR	SVM
Exploring Venus	S	1,862	0.125	0.118	1.000	1.000
Summer projects	I	1,750	0.047	0.045	0.990	0.992
Facial action coding system	S	2,167	0.030	0.030	0.990	0.994
Distance learning	I	2,157	0.029	0.024	0.986	0.990
Mandatory extracurricular activities	I	1,670	0.017	0.011	0.982	0.986
Does the electoral college work?	S	2,712	0.009	0.007	0.922	0.961
“A Cowboy Who Rode the Waves”	S	1,372	0.008	0.014	0.926	0.956
Seeking multiple opinions	I	1,552	0.006	0.006	0.983	0.986
Phones and driving	I	1,167	0.005	0.005	0.829	0.865
The Face on Mars	S	1,583	0.005	0.005	0.935	0.952
Car-free cities	S	2,665	0.005	0.003	0.987	0.990
Community service	I	1,542	0.003	0.001	0.989	0.991
Grades for extracurricular activities	I	1,626	0.001	0.001	0.967	0.974
Driverless cars	S	1,886	0.001	0.000	0.953	0.956
Cell phones at school	I	1,656	0.000	0.000	0.974	0.976
All 15 topics, pooled		27,367	0.020	0.019	0.972	0.980

Per topic. Most errors are false accusations, and for three of the four models, most of them come from *Exploring Venus* (Table 5). There, logistic regression, the SVM and DistilBERT flag between 10.5% and 15.7% of the student essays. ModernBERT is the most even model: its worst topic has an FPR of 1.9%.

All 15 topics. To check whether *Exploring Venus* is an accident of the split, I also hold out each of the 15 topics in turn and train the classical models on the other 14 (Table 6). *Exploring Venus* is the worst topic for both, with an FPR of 12.5% for logistic regression. It is not the only bad one: three other topics are above 2%, while the median topic is at 0.6%. On the three test topics of the topic split, the FPR of logistic regression only falls from 5.5% to 4.4%, so training on 14 topics instead of 10 helps a little but does not fix *Exploring Venus*.

6 Evaluation and Discussion

The near-perfect random-split scores are consistent with Alikhanov et al. (2026): when every topic is in both training and test, a model can lean on topic words. A teacher's new assignment looks like the topic split, so that is the number that matters.

6.1 False Accusations Depend on the Assignment

The FPR of about 5% is an average that hides one topic. With logistic regression, a class of 30 students who all wrote their own essay about Venus would see four or five of them flagged, and almost none on the other two topics.

I read four of the Venus essays that DistilBERT flagged, and all of them discuss how well the author of an article supports a claim. Source-based prompts as a group are not the problem, though. Without *Exploring Venus*, the other six source-based topics have a pooled FPR of 1.0% for logistic regression, against 1.5% for the eight independent ones, and the other bad topics in Table 6 include both kinds.

A better threshold would help: Recall@1% shows that the recall could stay between 94.3% and 99.6% at an FPR of 1%. But that threshold is set on the test scores, and one tuned on other topics may not transfer to a topic like *Exploring Venus*.

6.2 No Model Is Best on Every Topic

ModernBERT flags fewer students than DistilBERT on all three test topics, but my design cannot say why: the two differ in size, pretraining data and pooling. Truncation does not explain it: DistilBERT's FPR is 5.3% on the essays that fit within 512 tokens and 3.7% on the ones it cuts, although the two groups are not spread evenly over the topics.

ModernBERT is far better on *Exploring Venus*, but the classical models flag fewer students on the other two test topics together (Table 5). Since the transformers were tested on only three topics, I cannot say which model is safest on a new assignment.

6.3 What the Classical Models Learned

In the weights of logistic regression on the topic split, typical LLM phrasing such as "in conclusion" and "additionally" points to AI, and "because" is the strongest student signal.

Apostrophes. The default scikit-learn tokenizer drops single letters, so "it's important" and "you're" become "it important" and "you re", and both point to AI. "Dont", written without an apostrophe, points to student. So the model has partly learned that AI uses apostrophes correctly. A spell checker, which counts as AI-assisted editing (level 1) on the informal six-level scale of Monigatti (2025), might then push a student's own essay towards AI, which I have not tested.

Topic words. "Car", "cars" and "driving" point to student, a shortcut from the training topic *Phones and driving*, which has only 26% AI essays against 44% overall. *Driverless cars* has the highest miss rate for every model, though the difference is small (Table 5).

6.4 Missed AI Essays

All missed AI essays come from three of the ten sources in the test set: Llama 70B, Cohere Command and Mistral 7B Instruct v2. One that a transformer misses imitates a student's spelling mistakes: "I think student should have the rite to grade they're teachers".

6.5 Limitations

Each model is run once, with one seed and untuned transformer settings, and the transformers are tested on only three unseen topics. The confidence intervals resample essays, not topics, so they understate this uncertainty. Nearly all student essays come from one corpus, so a model can partly learn "this corpus" rather than "human". The essays are English school essays with binary labels, so Norwegian texts, newer language models and the middle levels of Monigatti's scale, such as co-writing, are not covered.

7 Conclusion and Future Work

The main lesson is that an average false positive rate says little about a single assignment. For logistic regression it was 15.7% on *Exploring Venus* and 0.1% on the other two test topics. Over all 15 topics, *Exploring Venus* is the worst but not the only bad one, and the bad topics include both kinds of prompt. A teacher cannot tell in advance whether a new assignment is one of them.

On a random split, all four models are almost perfect (RQ1, RQ2). On unseen topics, they still catch at least 97% of the AI essays, but three of them flag about 5% of the students (RQ3), against 0.9% for ModernBERT (RQ2). For three of the four models, most false accusations come from one topic, and all missed AI essays come from three generators (RQ4).

Future work could test the transformers on more topic splits, set the threshold on validation data, test edited, co-written and Norwegian essays, and add NeoBERT (Le Breton et al., 2025) as a third encoder.

Use of AI Tools

I used Claude (Anthropic) to help plan the project, write and explain the code, and draft this report. I ran all experiments myself, and every number comes from the result files in the repository.

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